

LIST CREATION DATE		2026-05-22				
Part name					Quantity	Optional?
01 PRIMARY PEDALS						
Pedal_GearBase_frame.stl					1	
Bearing_TopPlate.stl					1	
PedalCarriage.stl					4	
BarSleeve_CamRoller.stl					1	
BarSleeve_Magnet.stl					1	y
Base_FrontYoke.stl					1	
Base_RearYoke.stl					1	
BasicFootPedal.stl					2	y
Heel_Rest(Mirror).stl					1	y
Heel_Rest.stl					1	y
Rear_WallStiffenBar(Mirror).stl					1	y
Rear_WallStiffenBar.stl					1	y
WallStopperBase.stl					1	y
WallStopperBase(Mirror).stl					1	y
WallStopper_Slider.stl					1	y
WallStopper_Slider(Mirror).stl					1	y
Cam_rollerSpacer.stl					1	
Pedal_608_Bearing_Mount.stl					4	
02 Toe Brakes System						y
Toebrake_MountPlate.stl					2	
ToeBrake_Airbus_Texture.stl					2	y
ToeBrake_Boeing_Pedal.stl					2	
Toebrake_608_PillowBlock.stl					2	
Toebrake_608_PillowBlock(Mirror).stl					2	
ToeBrake_POT_BIG_GEAR.stl					2	
ToeBrake_Hall_Bracket.stl					2	
Toebrake_HalleffectMount.stl					2	
Toebrake_PedalStopBlock.stl			Note: For use with higher pedal angle position		2	y
TPU_DAMPER.stl			Note: Must be printed in TPU filament		2	y
Toebrake_SpringARm.stl					1	
Toebrake_SpringARm(mirror).stl					1	
ToeBrake_SpringBracket.stl					1	
ToeBrake_SpringBracket(mirror).stl					1	
Toebrake_4mm_CableRelief.stl					2	
2020_WIRE_RELIEF.stl					2	y
		02a Potentiometer Variant				y
		ToeBrake_POT_BRACKET.stl			2	y
		Toebrake_POT_BRACKET_CAP.stl			2	y
		Pot_6mm_SmallGear.stl			2	y
03 Electrical System						
PCB_Bracket.stl					1	
		Microcontroller Variants	(if not using pcb breakout board)			y
		ProMicro_Mount.stl			1	
		ProMicro_Mount.stl			1	
		Yaw Axis Sensing Variants	Dependant on which sensor setup you use			y
		AS5600_Sensor_Mount_vr3.stl			1	
		HallSensor_Mount.stl			1	
		TLE5011_frame.stl			1	
		Toplate_HalleffectMount.stl			1	
		03a Potentiometer Variant				y
		Pot_6mm_SmallGear.stl			1	
		Pot_BearingSleve_GearAddon.stl			1	
		YAW_Pot_Holder.stl			1	
04 Optional Covers						y
REAR_PCB_COVER.stl					1	
BottomCover.stl					1	
HallSensor_Strap.stl						
		USB Cover insert variants	Dependant on which microcontroller, cable orientation to use.			y
		PCB_PORT_INSERT_6mmWire.stl	for just a straight wire		1	
		PCB_PORT_INSERT_FJ_USB.stl	if using the USB B port on the breakout PCB and freejoy		1	
		PCB_PORT_INSERT_Top6mmWire.stl	cover oriented in a way that pushes the cable away from the main body		1	
		PCB_PORT_INSERT_SOLID.stl	solid one for the port you aren't using		1	
04 Piston Damper System						y
PISTON_DAMPER_FRONTBRACKET.stl					1	
PISTON_DAMPER_REARBAR.stl					1	

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PRIMARY HARDWARE		Notes			
1KG Spool PETG 1.75mm filament	2	PLA+ untested. Do not print the CAMs in PLA as they will distort over time. if using PET-CF make sure you get good layer adhesion			
50g TPU filament	1	If you are using the optional TPU_DAMPER.stl for the toe brakes.			
2020 extrusion 40cm lengths	4	One of them may be optionally cut in half for the heel rests or used straight			
M8x100 bolt	2	prefer hex head as they use a nylock nut and are easier to drive in with a socket wrench.			
M8x60 socket head bolt	4	Must be socket head, they recess into the part at the head and cannot work with hex head.			
M8x35 socket head bolt	9	You can also use hex head screws for these			
m8 nuts	10	get standard sizes, many of these are dropped into parts as internal threads and thus have to be nominal ISO sized. (~6mm thick, ~15mm widest corner to corner)			
m8 nylock lock nuts	6	get standard sizes, many of these are dropped into recesses and thus have to be nominal ISO sized. Not thinner than standard (~8mm thick, ~15mm widest corner to corner)			
m8 washer	6				
m8 spring washer	2				
m6x25 socket head bolt	4	Must be socket head, they recess into the part at the head and cannot work with hex head.			
m6 nylock lock nuts or flange nuts	4	You can use flange nuts instead of nylock nuts if the m6x25 doesn't have enough thread to bite the nylon part of the nut			
M5x50 Socket Head bolt	10				
M5x20 Socket Head bolt	30				
M5x20 COUNTERSUNK bolt	6	Countersunk, so angled top, measured from top face to end of threads 20mm.			
M5x16 Socket Head bolt	28				
M5x8 Socket Head bolt	17				
M5 nuts	43	get standard sizes, many of these are dropped into parts as internal threads and thus have to be nominal ISO sized. (~4mm thick, ~9mm widest corner to corner)			
M5 t slot drop in nuts	39	Get the kind that drops in as its easier than having to slide them in one after enough and forgetting one.			
m5 washer	4				
m4x20 socket head bolt	4				
m4x16 socket head bolt	16	Must be socket head, they recess into the part at the head and cannot work with hex head.			
m4 nuts	6				
m3x12 bolt	9	9th one is optional if using the potentiometer option for the gear teeth mount.			
m3x6 bolt	10				
608 skateboard bearings	20	Get good ones. Buy in a 20 pack.			
6003 bearings	2	same big bearings from the main gimbal.			
Extension spring with loop ends. ~55mm unloaded	4	two for the cams on the main pedal, one for each toe brake. On the cams you might use just 1 spring for less force.			
FRT-C2-301-G1 Rotary damper		these can now be a bit tough to come by due to tariffs and other stupidity, these are made obsolete if you have the motorcycle steering damper installed.			y
DAMPER ADD-ON		Notes			
M8x40 socket head bolt	1	Socket head, this one is shorter so it doesn't hit at the back			
M8x50 socket head bolt	1	Spocket head			
M8 nylock lock nuts	2	This needs to be a nylon locking nut so you can tighten the grip but the damper can still swing naturally.			
Steering damper	1	Get a quality one, check if for "stiction" after getting it and return if bad, you'll get annoyed quick if its sticky or squeaks			
ELECTRICAL		Notes			
B10K 10K Ohm Potentiometer 3Pin 15mm	3	This is if you want to make the pedals with simple pots and gear driven, recommended for first timers but you'll want to upgrade to hall effect later on			y
A1324LUA-T Hall Effect Sensor	3	This is for hall effect sensor option and is recommended. Get quality ones not the cheaper SS49E sensor which has bad noise at 3.3v with freejoy.			
1 meter 3 strand flexible wire	1	This is what runs to each pedal for the toe brakes, so something that can handle lots of flexing over time.			
8x3mm neodymium magnets	5	You need two per toe brake and one for the main pedals if not using the potentiometer config. Fridge magnet type with north/south of the flat faces.			
AS5600 sensor and breakout board	1	This is an alternative to the Hall effect sensor for the yaw axis, and is superior if you can burn the range using the extra script.			
Arduino Pro Micro	1	this is for MMJOY2 based controller		Either Or	
STM32F103C8 "Bluepill"	1	This is for Freejoy based controller. Note that there is a plague of counterfeits, only by from trusted seller.			
OSVBS breakout pcb	1	This is optional if you've had a set made from the gimbal you will have a spare. It allows use of the USB-B port with freejoy			y